Smart Plant Disease Detection

About & Mini Project Documentation

1. Project Overview

Smart Plant Disease Detection is an AI/ML-ready web application designed to help users analyze plant-leaf images and support early identification of possible plant diseases. The application was developed using Base44 AI with a simple, responsive and user-friendly interface.

The project demonstrates how Artificial Intelligence, Machine Learning, Computer Vision and web application development can be combined to address an agriculture-related problem.

2. Problem Statement

Farmers and plant growers may face difficulty identifying plant diseases at an early stage. Manual identification can require experience or expert assistance, which may not always be immediately available. Incorrect or delayed identification can affect plant health and crop productivity.

The proposed system provides a web-based workflow where a user can upload a plant-leaf image and receive a structured disease-analysis result. The application is designed to display the plant condition, predicted disease, confidence score, symptoms, possible causes and preventive measures when connected to a trained Machine Learning model.

3. Objectives

• Develop a simple and accessible plant disease detection web interface.

• Allow users to upload plant-leaf images for analysis.

• Create a structured interface for ML-based prediction results.

• Display disease-related information such as symptoms, causes and prevention.

• Create a foundation for integrating a real Computer Vision / Deep Learning model.

• Demonstrate a practical application of AI and web technology in agriculture.

4. Main Application Workflow

Upload Leaf Image → **Image Analysis** → **ML Model** → **Disease Prediction** → **Confidence Score** → **Symptoms & Prevention**

5. Key Features

Feature	Description
Image Upload	Upload a plant-leaf image for analysis.
Image Preview	Preview the selected image before analysis.
Prediction Interface	Designed to display plant health, disease and confidence.

Feature	Description
Disease Information	Displays symptoms, possible causes and preventive measures.
Prediction History	Can store and display previous analysis results.
Dashboard	Can summarize analyses and prediction statistics.

6. Technologies Used

Technology	Purpose
Base44 AI	AI-assisted web application development
HTML / CSS / JavaScript	Web interface and application foundation
Python	Planned Machine Learning model development
TensorFlow / PyTorch	Planned Deep Learning implementation
OpenCV	Planned image processing
CNN / Transfer Learning	Planned plant disease image classification
GitHub	Version control and project documentation

7. Machine Learning Integration

The current Base44 application is structured to support integration with a real Machine Learning model. The intended ML pipeline is:

Plant Disease Dataset → Image Preprocessing → Training → Model Evaluation → Saved Model → Prediction API → Base44 Web Application

A CNN or transfer-learning model can be trained on labeled plant-leaf images. The trained model can then be exposed through a Python API and connected to the web application.

8. Current Project Status

Component	Status
Base44 web interface	Completed
Image upload workflow	Implemented
Prediction-result interface	Implemented
ML model training	Planned / In progress
ML prediction API	Pending
Live ML prediction	Pending integration

9. Future Enhancements

• Integrate a trained CNN or transfer-learning model.

• Support multiple plant species and disease categories.

• Add real-time camera/image capture.

• Improve model accuracy using larger and better-quality datasets.

• Add multilingual support for wider accessibility.

• Add model performance and prediction analytics.

• Deploy the ML prediction API and connect it to the live website.

10. Learning Outcomes

This mini-project provides practical exposure to AI-assisted web development, Machine Learning project planning, Computer Vision, Deep Learning concepts, image classification, API integration, GitHub and real-world AI application development.

11. Project Link

Live Application: https://lean-flora-scan-pro.base44.app

12. Author

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13. Disclaimer

This project is developed for educational and demonstration purposes. Any future ML prediction should be treated as informational and should not replace professional agricultural diagnosis or advice.